2 SOFTWARE

The programs that run on a computer are called software. Programs are instructions that are carried out or executed by the computer's processor, which provides other instructions for the rest of the computer.

Some computers can only run one program at a time, while others can run multiple programs at the same time. Some computers are single purpose, which means that they are only able to run one piece of software (for example, the software used to control a microwave). Other computers are general purpose, which means that they are able to carry out a range of tasks (for example, the software used in a tablet computer).

There are two basic groups of software: system software and application software.



LEARNING OBJECTIVES

- Be able to identify the purpose of systems software and applications software
- Know about operating systems and system software tools such as utilities
- Know about the role/function of the operating system, including basic knowledge of:
  - single user and network
  - memory management
  - resource management
  - security
  - print spooling
- Know about software applications (apps), including office-productivity tools, web authoring, image and sound editing, presentation software, control software, project management software
- Know about software licensing types (free/open source and proprietary software)
- Understand that the purpose of communication software is to provide remote access to systems and to exchange files and messages in text, images, audio and/or video formats between different computers or users
- Know why software is updated, how it is done and possible risks to data/systems
- Understand that settings of ICT systems can be configured to meet the accessibility needs of individuals
- Be able to justify choices made in identifying and configuring hardware and software

SYSTEM SOFTWARE

Programs that are designed to maintain or operate the computer system are known as system software.

UTILITY SOFTWARE

Utility software is one form of system software, which carries out configuration and maintenance tasks.

SUBJECT VOCABULARY

utility software (also known as utilities) system software that carries out configuration and maintenance tasks

BACKUP

Backup utilities create a copy of files and programs. Backups can be set to run automatically (usually at a time when the system is not in use) or can be started by a user.

You will learn more about backups in *Unit 3 Operating online* (pages 104–105).

DEFRAGMENTATION

GENERAL VOCABULARY

fragmented broken into pieces

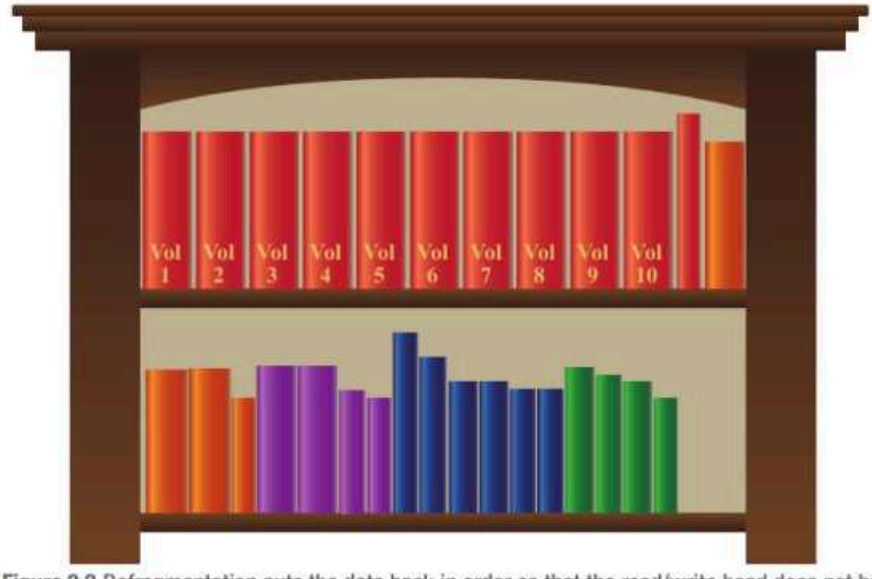
As data is stored to hard disk, some systems spread it across the disk wherever there is free space. This means that sometimes data is **fragmented** and stored out of order or out of sequence.



▲ **Figure 2.1** Data is sometimes stored in the nearest free space, which results in data getting out of sequence. This is an illustrative example.

Data from the same file may be spread over multiple locations on a platter, on different platters in the cylinder or sometimes even on different disks. When these files are loaded, data has to be retrieved from these multiple locations. The read/write head of the hard drive has to find this data, which takes extra time and slows down the process of loading the files.

To avoid this and to speed up the system, disk defragmentation utilities reorder the fragments of data so that they are stored as close to each other as possible.



▲ **Figure 2.2** Defragmentation puts the data back in order so that the read/write head does not have to search the disk for fragments of data. This is an illustrative example.

Defragmentation utilities are usually set to run automatically, but they can also be started by a user.

SKILLS EXECUTIVE FUNCTION
INTELLECTUAL INTEREST AND CURIOSITY
SELF-DIRECTION

ACTIVITY

▼ DEFRAGMENTATION

1 Run a defragmenter analysis on your system. When was the analysis last run? How fragmented was the data last time the analysis ran?

2 Research the need for defragmentation on SSDs.

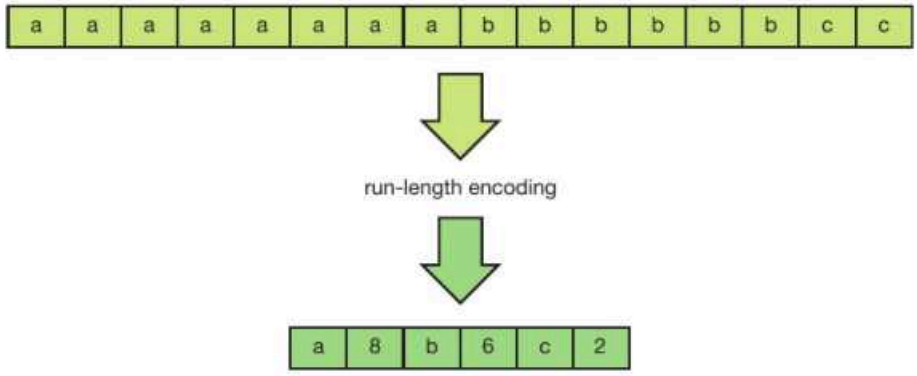
COMPRESSION



▲ **Figure 2.3** Data can be compressed to remove unnecessary use of code or space in the file. This is an illustrative example.

Compression utilities reduce the original size of a file or set of files.

Where there are repeated patterns of data, rather than storing every repeated instance, only the first instance of the data is stored, alongside how many times it is repeated. For example, in Figure 2.4, there were eight instances of a, six instances of b and two instances of c in the original file. This data can be compressed to be stored as 'a8b6c2', which reduces the amount of storage needed.



▲ **Figure 2.4** Run Length Encoding is one method of data compression

SUBJECT VOCABULARY

decompress extract data from a compressed file so that it can be read

The resulting compressed file is not usually readable by the original application. Compression utilities also have the ability to **decompress** (extract) the data from a compressed file so it can be read by the original application again.

FORMATTING

Disk formatting prepares storage media such as a hard disk drive or USB flash drive for its first use. If a disk has already been used, then formatting it will make all of the data on the disk unreadable by normal applications.

DID YOU KNOW?

Different operating systems require disks to be formatted in certain ways in order to perform specialist tasks. Disks can also be formatted in a way that makes them compatible with all operating systems.

Disk formatting utilities make this process easy for users by providing them with the option to choose the file system, file unit sizes and the name of the disk. Perhaps most importantly, they make sure that the user really wants to format the disk.



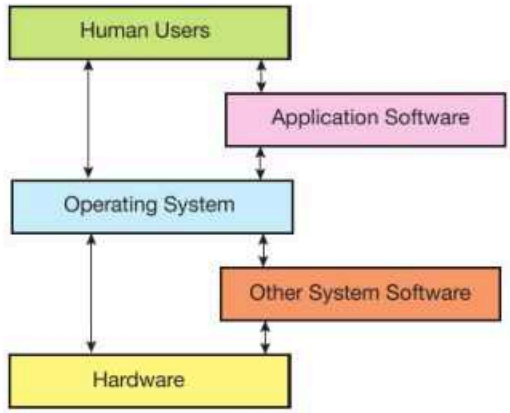
▲ **Figure 2.5** Accidentally formatting the wrong disk could result in the unwanted loss of data

DID YOU KNOW?

Formatting a disk will not erase the data completely and it can still be recovered using specialist tools, but there are some utilities which will overwrite the data so that files cannot be recovered.

OPERATING SYSTEMS

The operating system (OS) allows the user to control and manage the computer's hardware.



▲ **Figure 2.6** The operating system lets the user control the hardware, either directly or by using applications software or system software

GENERAL VOCABULARY

customise to change something in order to make it more suitable for a particular user or to make it look special or different from things of a similar type

Single-user operating systems only allow for a general user and do not provide the option to **customise** the user interface for different users. They are often found in household appliances.

Network operating systems have additional functionality, including:

- sending requests to a server when users log in with their username and password
- separating user accounts and ensuring that users cannot access each other's files
- providing access to network storage and shared resources such as networked printers.

MEMORY MANAGEMENT

The operating system allocates the required amount of memory (RAM) to one or more applications. When the application no longer requires the memory space, the operating system makes the space available for other applications to use.

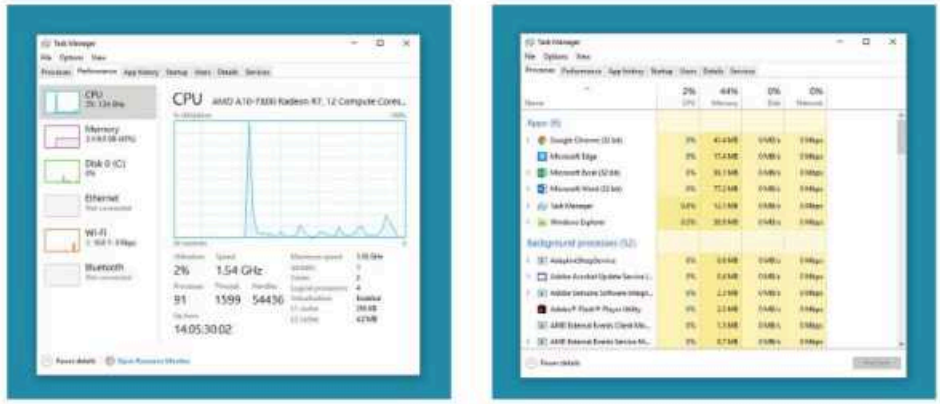
The operating system also handles the creation of virtual memory. See page 60 for more information about virtual memory.

RESOURCE MANAGEMENT

SUBJECT VOCABULARY

graphics card a device built into or added into a computer to allow it to display visual graphics

System resources include internal components, such as the processor and **graphics card**, and external devices, such as printers. As well as allocating memory, the operating system ensures that system resources are made available to applications when they are required. If a system resource is already being used by another application, then the operating system may say that the required resource is in use and place the additional request in a queue. For some resources, such as printers, the operating system can tell the user when the resource becomes available.



▲ **Figure 2.7** Information about the use of memory and resources is provided by utility software

SECURITY

SUBJECT VOCABULARY

authentication the process of confirming that a user is permitted to access certain files, hardware and software

firewall a system (hardware or software) that protects a computer network from being used or looked at by people who do not have permission to do so

Operating systems can give users secure access to a computer's storage and other hardware through usernames and passwords, biometric scanning (see pages 27–28) or personal access cards. This process is called **authentication**. Operating systems can provide software **firewalls** to authorise or prevent network data from remotely accessing a service or application.

PRINT SPOOLING

SUBJECT VOCABULARY

print spooling the process of keeping pages queued in order, ready to be printed by a printer

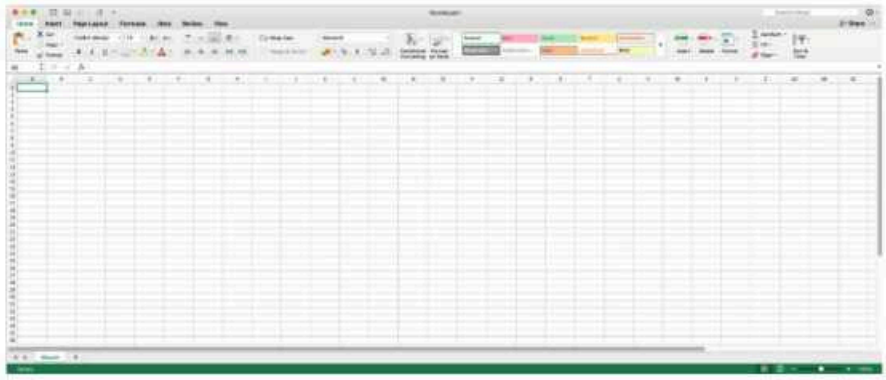
During large print jobs, the computer will have the pages ready for the printer faster than the printer can produce them. The operating system keeps each page in a queue ready for printing. This process is called **print spooling**.

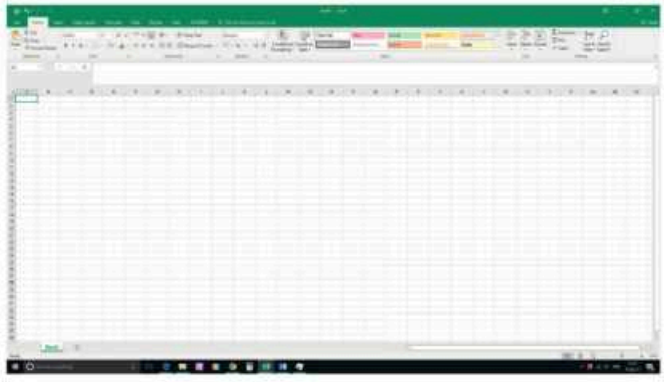
APPLICATION SOFTWARE

Software applications (apps) allow users to produce a digital product, such as a presentation or image, or carry out specific tasks that are not related to the operating system.

Apps are usually downloaded from servers on the internet. Some devices, like smartphones and tablets, use operating systems that provide access to app stores. Some of these only make apps available to users after the store has checked the quality of the app and has ensured that the app does not include code that will harm a user's device or data. Apps can also be installed from storage media like DVDs.

You will learn more about online software in *Unit 4 Online goods and services* (178–179).





▲ **Figure 2.8** Some developers provide versions of their apps for different types of computer and operating systems. The top window shows the Apple macOS version whilst the bottom window shows the Microsoft® Windows version.

HINT

When you are writing about application software in your exam, you should refer to the application by its type rather than by a particular name. For example, use the term 'word processing software' rather than the brand name Microsoft® Word.

You need to be able to identify the type of software that meets specific needs. Each type of application software has particular features that make it suitable to perform a particular function. Sometimes more than one type of application can complete the task, but you should consider which one is most suitable.

OFFICE PRODUCTIVITY SOFTWARE

SUBJECT VOCABULARY

application suite a collection of application software that share the same look and user interface; often, they can share data between each other and share some functionality
hyperlink a link that can be clicked in order to go to another location (often a web page on the internet)

Office productivity software is any application that can carry out work-related tasks. They are often available as an **application suite**.

WORD PROCESSING

Word processing software lets users create documents that mostly include words, but also some images. They can also include tables, **hyperlinks**, equations, simple drawings and shapes and charts. An example of word processing software is Writer from the LibreOffice application suite.

Word processors are a good choice of application for creating:

- letters
- reports
- essays
- books (for example, this book was written using word processing software).

Some word processing applications allow people to work together on a document. Comments can be added to the text so that others can understand the writer's intention and give feedback. The software can also track any changes made to the document, giving others the option to accept or reject those changes.

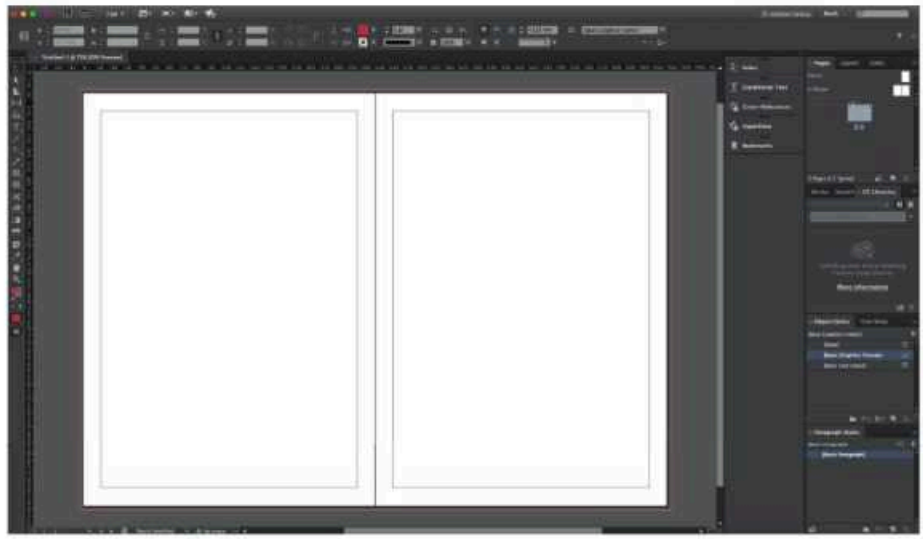
SUBJECT VOCABULARY

font the size and style of text

Word processing applications allow users to change the **font** and provide different layout views to allow users to position text and images on the page. They usually come with spelling and grammar checking tools that identify errors in the text as the user types.

DESKTOP PUBLISHING (DTP)

DTP software has many similar features to those in word processing software. The main difference is the way in which the software allows users to work with complicated page layouts. An example of DTP software is Adobe® InDesign®[2].



▲ **Figure 2.9** Page layouts in DTP software help the designer to see what the final version will look like when printed

SPREADSHEET

Spreadsheet applications are used to do calculations. Users can use and create formulae and functions to perform automatic calculations on values that can be entered and changed later. These features allow spreadsheets to be used to model financial scenarios and answer 'what if?' questions, such as '**What** would the cost per ticket be **if** I changed the maximum number of staff required at my event?'.

Venue	Gold	1		3000
Catering package	Gold	1		30
Entertainment	Bronze	3		300
Decorations	Silver	2		300
Transport	Gold	1		80
Number of Staff required				20
Ticket price needed			£	59.25
Meets requirements?				No

Venue	Gold	1		3000
Catering package	Gold	1		30
Entertainment	Bronze	3		300
Decorations	Silver	2		300
Transport	Gold	1		80
Number of Staff required				30
Ticket price needed			£	55.92
Meets requirements?				No

▲ **Figure 2.10** Two scenarios from a financial model

Venue	Gold	1	=VLOOKUP(C11,$L$3:$R$5,5)
Catering package	Gold	1	=VLOOKUP(C12,$L$3:$R$5,3)
Entertainment	Bronze	3	=VLOOKUP(C13,$L$3:$R$5,4)
Decorations	Silver	2	=VLOOKUP(C14,$L$3:$R$5,6)
Transport	Gold	1	=VLOOKUP(C15,$L$3:$R$5,7)
Number of Staff required			=ROUNDUP(D5/10,0)
Ticket price needed			=(G14-J14)/D5
Meets requirements?			=IF(D19>D8,"No","Yes")

▲ **Figure 2.11** Formulae and functions are used for calculations that are carried out automatically when values change

2 ADOBE AND ADOBE INDESIGN ARE EITHER REGISTERED TRADEMARKS OR TRADEMARKS OF ADOBE SYSTEMS INCORPORATED IN THE UNITED STATES AND/OR OTHER COUNTRIES

An example of spreadsheet software is Microsoft® Excel®.

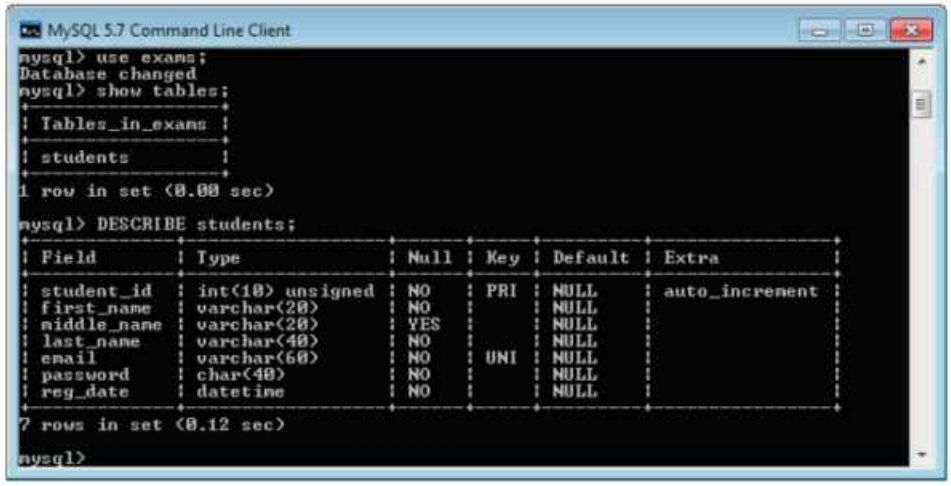
You will learn more about spreadsheets in *Unit 5 Applying Information and Communication Technology* (page 185) and *Unit 6 Software skills* (page 276).

DATABASE

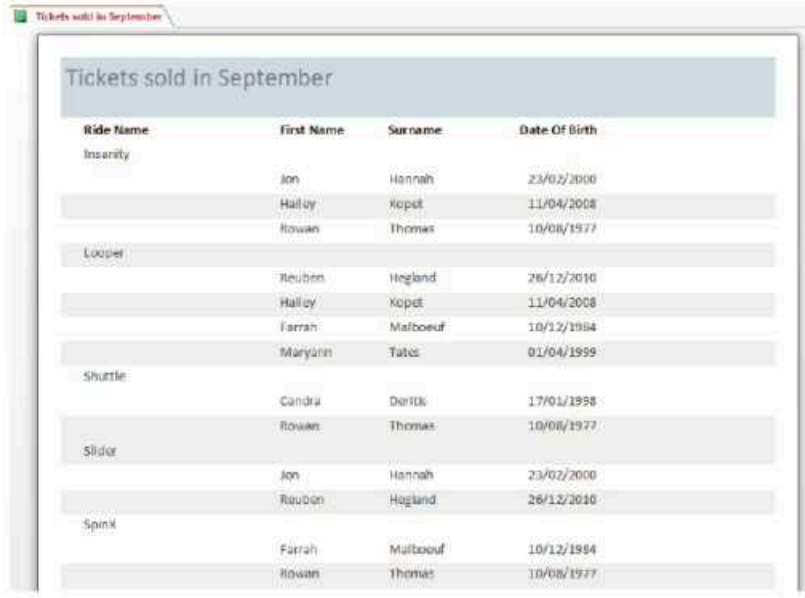
Database management systems (DBMS) are used to enter, edit and search data. Some systems can also produce reports that dynamically display real-time changes to the data as it updates.

DBMS software features include:

- data entry forms so that users can input data
- query editors so that users can select all data that meets certain criteria
- report builders so that users can display data in a more readable format.



▲ **Figure 2.12** This DBMS software uses a command line interface and contains fewer features


Tickets sold in September

Ride Name	First Name	Surname	Date Of Birth
Insanity			
	Jon	Hannah	23/02/2000
	Hailey	Kopet	11/04/2008
	Rowan	Thomas	10/08/1977
Looper			
	Reuben	Hegland	26/12/2010
	Hailey	Kopet	11/04/2008
	Farrah	Malboeuf	10/12/1984
	Maryann	Tates	01/04/1959
Shuttle			
	Candra	Dertik	17/01/1958
	Rowan	Thomas	10/08/1977
Slider			
	Jon	Hannah	23/02/2000
	Reuben	Hegland	26/12/2010
SpinX			
	Farrah	Malbouf	10/12/1984
	Rowan	Thomas	10/08/1977

▲ **Figure 2.13** A report from a database

An example of DBMS software is Oracle® MySQL.

You will learn more about databases in *Unit 6 Software skills* (pages 307–333).